

1. Application Introduction

Welcome to **Air Taxi eVTOL Pulse**. This app is built with Qt Quick and C++ to show real-time eVTOL flight data.

It receives live data using MQTT. If MQTT is offline, the app switches to a built-in flight simulation mode.

2. Main Header

The top header shows your overall mission status:

- **Left:** Current date, local time, and total flight time.
- **Centre:** The Air Taxi operations dashboard title, plus an MQTT panel that tracks your connection, battery level, motor temperature, GPS, message queue, and dropped packets.
- **Right:** A moving trajectory animation showing speed, altitude, and path.

3. Primary Flight and Navigation

This page features six interactive instruments.

- **Airspeed:** Shows calibrated airspeed (for flying the plane) and true airspeed (how fast you are actually moving).
- **Altitude:** Combines barometric altitude (flight level) with radio altitude (your exact height above the ground).
- **Vertical Speed:** Displays your climb or descent rate in feet per minute, changing colour and direction based on your movement.
- **Attitude Indicator:** Shows pitch, nose angle, side-to-side bank, and rotation.
- **Heading and Track:** Compares your magnetic heading with your actual ground path to track navigation drift.
- **Flight-Path Vector:** Projects your planned path, route quality, and deviation.

4. Distributed Electric Propulsion

This section monitors the entire electric motor network:

- **Propulsion Map:** An animated diagram showing power moving from the main battery to all four motors, including individual and average RPMs.
- **Vector and Thermal Control:** Tracks nacelle tilt (shifting between vertical lift and forward flight) and motor heat to prevent overheating.
- **Thrust Allocation:** Compares front and rear power groups to verify that total thrust is balanced.
- **Inverter Control:** Monitors voltage, current, power, and health for every motor channel to spot failing parts quickly.

5. Energy, Power, and Battery

This module tracks your high-voltage power storage and distribution:

- **Battery State of Charge:** Shows current charge, usable capacity, and remaining run time, warning you when reserves get low.
- **Cell Thermal Map:** Displays module temperatures, showing minimums, maximums, and spreads to catch overheating early.
- **Power Consumption:** Tracks total power demand, separating hover mode (which takes a lot more power) from cruise mode.
- **Battery Health:** Compares current capacity to a brand-new battery and tracks overall degradation.
- **High-Voltage Bus:** Distributes power to motors, electronics, and safety systems.

6. Flight Control System and Automation

This page manages aircraft steering, automation, and movement:

- **Flight Modes:** Switch between hover, transition, cruise, and vertical landing. Choose automatic mode to follow the flight plan, or manual mode to pick a specific phase.
- **Autopilot:** Toggle between automated flight and manual fly-by-wire controls.
- **Control Authority:** Shows how much control the computer currently holds.
- **Actuator Deflection:** Displays left and right elevons, ruddervator, and nacelle positions. In manual mode, you can drag these controls to test fly-by-wire commands.
- **Route Management:** Shows the complete path between vertiports, including waypoints and direct-to navigation options.

7. Safety, Health, and Environment

This final module handles environmental safety, navigation locks, and landing guides:

- **Acoustic Noise:** Estimates noise using motor speed and airspeed, warning you if you go over community sound limits.
- **Ballistic Parachute:** Manages the emergency recovery parachute. It must be armed first, and will only deploy if altitude and airspeed are safe. A test override is also included.
- **Collision Avoidance:** Tracks nearby traffic, bearing, and distance, with adjustable radar range.
- **GPS and GNSS Integrity:** Shows satellite count, signal strength, and navigation accuracy, with a test mode to simulate bad signals.
- **Vertiport Landing Guidance:** Helps with precision landings by showing lateral and vertical alignment, radar height, and approach modes.

8. Closing

Air Taxi eVTOL Pulse brings flight, power, navigation, and safety tools together into a single dashboard. it runs smoothly on simulated test data or live aircraft telemetry.

This is a personal development project, and the underlying code is available in my personal Git repository.

Thanks